SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai – 602 105

CO3 Scenario Based Assignment: Design and Performance Evaluation of an Integrated Photodetection, Imaging, Biomedical Sensing and Display System

ECA1804 – OPTOELECTRONICS AND OPTICAL COMMUNICATION

Submitted by

Field	Details
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Course	ECA1804 – Optoelectronics and Optical Communication
Course Outcome	CO3
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Date of Submission	13-October-2026

Summary of the solution: Use a Si PIN photodiode for PPG and imaging, an InGaAs APD (M ≈ 10) at 1550 nm for LiDAR, a CMOS image sensor with NIR illumination for face recognition, an AlGaN/SiC PIN for UV, flexible PV for auxiliary power, and a MicroLED (AR/VR) / OLED (foldable) display.

Key result	Value
PPG photocurrent variation	15 µA
LiDAR photocurrent (PIN / APD M = 10)	2.4 µA / 24 µA
Low-light SNR	15 dB (ratio 31.6)
Transit-time bandwidth	22 GHz
PV maximum power	2.25 W

Stage 1 – Interpreting the application requirements (15 marks)

Each sub-system has a different main need: PPG needs a sensitive, linear and low-power detector; LiDAR needs speed and gain; imaging needs small low-noise pixels; the display needs brightness and efficiency.

Requirement	What it means for the design
Wearable PPG	Detect small (µA-level) changes on a large DC level; high responsivity, linear, low noise, µW power
Automotive LiDAR	Weak, ns-wide return pulses; high gain (APD), GHz bandwidth, low noise, eye-safe wavelength
Weak-signal detection	Low dark current, low NEP, high detectivity D*
Responsivity, QE, D*	$R = \eta q\lambda/hc$ converts optical power to current; QE is the fraction of photons collected; D* compares detectors of different area
NEP	Smallest detectable power (SNR = 1 in 1 Hz); lower NEP = better low-light detection
Linearity	Photocurrent must stay proportional to power so PPG amplitude and pixel grey levels are correct
Speed	Set by transit time (carrier drift across depletion layer) and RC time constant (capacitance × load)
CMOS/CCD sensors	High fill factor, low read noise, high dynamic range, NIR sensitivity, on-chip processing (SoC)
Pixel scaling	Smaller pixels give higher resolution but collect fewer photons (lower SNR)
Flexible PV	Light, bendable, efficient under indoor light, safe on skin
OLED / MicroLED / QD	Brightness, contrast, response time, lifetime, power; AR/VR needs very high brightness and pixel density; foldables need flexibility
Practical constraints	Battery power, heat, size, reliability, cost and system complexity

Application Requirement Statement

> *The platform shall detect a 25 µW PPG variation with a linear low-noise detector, receive 3 µW-level ns LiDAR returns with high gain and GHz bandwidth, achieve an SNR well above 15 dB in low light, provide high-resolution low-power imaging for face recognition and automotive use, detect UV reliably, harvest auxiliary power from flexible PV, and drive a bright, efficient AR/VR and foldable display — all within wearable/automotive power, thermal and reliability limits, verified by MATLAB/SCILAB simulation.*

Stage 2 – Photodetector selection (20 marks)

The parameter that converts optical power to photocurrent is the **responsivity R (A/W)**: $I_p = R \times P$.

PPG photocurrent

$\Delta I_p = R \times \Delta P = 0.60\ A/W \times 25\ \mu W = 15\ \mu A$

In PPG, an LED lights the skin and the detector receives light reflected or transmitted through tissue. Each heartbeat increases blood volume, which absorbs more light, so the detected current has a large steady (DC) part and a small pulsing (AC) part. The 15 µA AC variation follows the pulse: its period gives heart rate, and the ratio of red/IR AC-to-DC values gives SpO_2.

LiDAR photocurrent

$I_p = R \times P = 0.80\ A/W \times 3\ \mu W = 2.4\ \mu A$ (PIN, M = 1)

With APD gain M = 10: $I = M \times R \times P = 10 \times 2.4\ \mu A = 24\ \mu A$

APD gain multiplies the signal before the amplifier adds its thermal noise, so weak returns rise above the receiver noise floor. At a weaker, more realistic 100 nW return the PIN SNR is −1.1 dB, while an APD at its optimum gain (M ≈ 14) gives 16.4 dB.

Quantum efficiency

$\eta = R \times hc / (q\lambda) = R \times 1.24 / \lambda(\mu m)$

- PPG (Si, assumed λ = 940 nm): $\eta = 0.60 \times 1.24 / 0.94 =$ **0.79 (79 %)**
- LiDAR (InGaAs, assumed λ = 1550 nm): $\eta = 0.80 \times 1.24 / 1.55 =$ **0.64 (64 %)**

R = 0.8 A/W at 905 nm would require η > 100 %, so the LiDAR detector must be InGaAs at 1550 nm (also the eye-safe band). Higher η means more photons give current, directly improving sensitivity.

PIN vs APD

Parameter	PIN photodiode	APD
Responsivity	0.5–0.9 A/W	R × M (e.g. 8 A/W at M = 10)
Internal gain	None (M = 1)	10–100
Sensitivity	Moderate	5–15 dB better
Noise	Shot + thermal; no excess noise	Excess noise factor F(M)
Bandwidth	Very high (GHz–tens of GHz)	High, limited by gain-bandwidth product
Bias voltage	0–10 V	30–60 V (InGaAs), 100–200 V (Si)
Linearity	Excellent over 6–8 decades	Gain varies with bias and temperature
Complexity / cost	Low	High-voltage supply,

Parameter	PIN photodiode	APD
		temperature compensation

Dark current, NEP and detectivity

Dark current flows with no light. It adds shot noise and an offset that can hide weak signals; it doubles roughly every 8–10 °C.

NEP = √(2qI_d) / R (shot-noise limited)
D* = √A / NEP

Detector (assumed)	I_d	Area	NEP	D*
Si PIN (PPG)	1 nA	1 mm²	$\sqrt{(2 \times 1.6\times10^{-19} \times 10^{-9})/0.6} = 2.98\times10^{-14}$ W/√Hz	3.35×10^{12} cm·√Hz/W
InGaAs APD (LiDAR, M = 10, F = 5.55)	10 nA	200 µm dia.	$\sqrt{(2qI_dF)/R} = 1.67\times10^{-13}$ W/√Hz	1.06×10^{11} cm·√Hz/W

Lower NEP and higher D* mean better weak-light detection.

Linearity

A PIN is linear from nW to mW, so PPG waveform shape and image grey levels are preserved. APD gain compresses at high power, so APDs suit pulse detection (LiDAR) rather than intensity measurement.

Design Decision 1

> ***I recommend a Si PIN photodiode for PPG and imaging, and an InGaAs APD (M ≈ 10, adjustable 5–15) for LiDAR and low-light detection, because*** *the PIN gives a linear 15 µA PPG signal with 79 % QE and a low NEP of 2.98×10^{-14} W/√Hz at low bias and power, while the APD raises the 2.4 µA LiDAR current to 24 µA and improves SNR from 28.2 to 34.1 dB at 3 µW (and by 17.5 dB at 100 nW) with GHz-class speed.*

Stage 3 – Noise and SNR strategy (15 marks)

The metric that compares signal with noise is the **signal-to-noise ratio (SNR)**.

SNR (dB) = P_signal (dBm) − P_noise (dBm) = −22 − (−37) = 15 dB

SNR (ratio) = 10^(15/10) = 31.6 (P_s = 6.31 µW, P_n = 0.20 µW)

An SNR of 15 dB is workable for PPG trend detection but marginal for reliable low-light pulse detection, where ≥ 20 dB is targeted.

Noise sources

Noise	Origin	Mean-square current

Noise	Origin	Mean-square current
Shot noise	Random arrival of photons/carriers	$i^2 = 2qI_pB$
Dark-current noise	Random thermal generation of carriers with no light	$i^2 = 2qI_dB$
Thermal noise	Random electron motion in load resistor	$i^2 = 4kTB/R_L$
Avalanche noise	Random multiplication in APD	$i^2 = 2q(I_p + I_d)M^2F(M)B$, $F = kM + (1 - k)(2 - 1/M)$

Effect of APD gain: signal grows as M^2 but shot noise grows as $M^2F(M)$. At low gain the SNR rises because thermal noise is overcome; above an optimum gain the excess noise F(M) dominates and SNR falls. For the InGaAs APD (k = 0.45, B = 500 MHz, R_L = 1 kΩ), optimum M ≈ 4.5 at 3 µW and ≈ 14 at 100 nW.

Dominant noise in low light: thermal (amplifier) noise and dark-current noise, because the signal shot noise becomes very small.

Falling optical power with constant dark current: signal power falls as P^2, shot noise only as P, and dark/thermal noise stays constant. SNR therefore drops by 20 dB per decade of power in the thermal/dark-limited region, until the signal is lost.

Bandwidth: every noise term is proportional to B, so halving B improves SNR by 3 dB. B should be just enough for the signal (≈ 0.35/rise time).

SNR improvement methods

Method	Effect
Optimum APD gain (M ≈ 10)	Overcomes thermal noise: +5.9 dB at 3 µW, +17.5 dB at 100 nW
Narrow optical band-pass filter	Removes sunlight/ambient background shot noise
Low dark current (small area, InGaAs/Si quality)	Lowers noise floor and NEP
Cooling (TEC)	Dark current halves for about every 8–10 °C reduction
Bandwidth optimization	+3 dB for each halving of B
Better optical coupling (lens, AR coating)	More signal power; SNR rises up to 20 dB per decade

Design Decision 2

> ***I recommend an APD at optimum gain (M ≈ 10, temperature-compensated bias) with a narrow optical band-pass filter, matched receiver bandwidth and a low-noise TIA as the noise/SNR configuration because*** *the given 15 dB SNR (ratio 31.6) is marginal; in*

> *low light the noise is dominated by thermal and dark-current noise, which APD gain overcomes until avalanche excess noise takes over; combining gain (+6 dB), bandwidth matching (+3 dB) and filtering lifts the SNR to above 20 dB.*

Stage 4 – Response time and bandwidth (10 marks)

The parameter linking transit time to bandwidth is **$f \approx 0.44 / t_r$**.

$f_{transit} = 0.44 / t_r = 0.44 / (20 \times 10^{-12}\ s) = 2.2 \times 10^{10}\ Hz = 22\ GHz$

Transit time: carriers generated by light must drift across the depletion layer (width W) at saturation velocity v_s; $t_{tr} \approx W/v_s$. A thicker layer absorbs more light (higher QE) but is slower.

RC time constant: the junction capacitance $C_j = \varepsilon A/W$ and the load R_L form a low-pass filter, $f_{RC} = 1/(2\pi R_L C_j)$. Larger area or thinner layer → more capacitance → lower bandwidth.

Example (assumed C_j = 1 pF, R_L = 50 Ω): **$f_{RC} = 1/(2\pi \times 50 \times 10^{-12}) = 3.18\ GHz$**. Combined:

$f_{3dB} = 1/\sqrt{(1/f_{tr}^2 + 1/f_{RC}^2)} = 1/\sqrt{(1/22^2 + 1/3.18^2)}\ GHz \approx 3.15\ GHz$

So the RC limit, not the transit time, sets the real bandwidth.

Larger active area gives	Smaller active area gives
Easier optical coupling, more collected light (sensitivity)	Less capacitance, higher bandwidth
More capacitance and dark current, lower bandwidth	Harder coupling (needs focusing lens)

For LiDAR: a 1 ns pulse needs B ≈ 0.35/1 ns ≈ 350 MHz, and 15 cm range resolution needs ~1 ns timing, so a 200 µm InGaAs APD (≈ 1 pF) with a lens and a TIA of ~0.5–1 GHz is sufficient. **For imaging:** pixel photodiodes integrate charge over ms frames, so speed is set by readout, not by the diode.

Receiver circuit: a transimpedance amplifier (TIA) holds the diode at virtual ground, reducing the effective RC; overall bandwidth is set by the slowest of diode, TIA and filter.

Design Decision 3

> ***I recommend a small-area (≈ 200 µm) InGaAs APD with a focusing lens and a low-noise TIA (≈ 1 GHz) as the high-speed detector configuration because*** *the 22 GHz transit-time bandwidth is far above need, the real limit is the RC bandwidth (≈ 3.15 GHz at 1 pF), which still exceeds the ≈ 350 MHz required for 1 ns LiDAR pulses; the lens recovers the coupling lost by the small area, keeping both speed and sensitivity.*

Stage 5 – PV, image-sensor and display technologies (15 marks)

A. Solar photovoltaic analysis

$P_{max} = FF \times V_{oc} \times I_{sc} = 0.75 \times 0.60\ V \times 5\ A = 2.25\ W$

- **V_{oc}** is the voltage at zero current (right end of the I–V curve); **I_{sc}** is the current at zero voltage (left end). Together they set the maximum possible power $V_{oc} \times I_{sc}$ = 3.0 W.

- **Fill factor** (= P_max / V_oc I_sc) measures how "square" the I–V curve is; series resistance, shunt leakage and recombination lower it.
- **Efficiency** η = FF × V_oc × I_sc / P_in. Assuming a 156 cm² cell under 1000 W/m² (P_in = 15.6 W): η = 2.25 / 15.6 = **14.4 %**. A higher FF gives a proportionally higher efficiency.

Flexible PV for wearables	
Advantages	Light, bendable, conforms to wrist/strap, works under indoor light (organic/perovskite/a-Si), extends battery life
Limitations	Lower efficiency (5–15 %), small area gives only mW–tens of mW, degradation from moisture and bending, needs a power-management IC and storage

B. CMOS vs CCD image sensors

Feature	CMOS	CCD
Architecture	Active pixel: photodiode + amplifier in each pixel	Photodiode pixels; charge shifted to one output amplifier
Pixel operation / readout	Each pixel converted to voltage, row/column addressed, on-chip ADC	Charge transferred pixel-to-pixel, read serially
Power	Low (tens–hundreds of mW)	High (multiple clock voltages)
Speed	High (parallel readout, > 100 fps)	Lower (serial readout)
Noise	Very low with correlated double sampling and BSI	Very low, uniform
Pixel scaling	Down to ≈ 0.6–1 µm with back-side illumination	Limited by charge-transfer efficiency
SoC integration	ADC, ISP and AI can be on the same chip	Separate support chips needed
Face recognition	Excellent (NIR-enhanced, low power)	Possible, but high power
Automotive	Excellent (HDR, AEC-Q100 qualified)	Rarely used

Smaller pixels raise resolution and reduce chip size and cost, but each pixel collects fewer photons, so SNR, full-well capacity and dynamic range fall, and crosstalk rises. Back-side illumination, deep trench isolation and pixel binning compensate.

C. OLED vs MicroLED vs quantum-dot displays

Feature	OLED	MicroLED	Quantum dot (QD-LCD / QD-OLED)
Light generation	Organic electroluminescence (self-emissive)	Inorganic GaN micro-LEDs (self-emissive)	QDs convert blue light to pure red/green
Brightness	≈ 1000 nits	> 1 000 000 nits (micro-display)	≈ 1000–2000 nits
Contrast	Infinite (true black)	Infinite	High (QD-OLED) / limited (QD-LCD)
Response	≈ µs	ns	ms (LCD) / µs (QD-OLED)
Efficiency / power	Good	Best	Good
Colour	Wide gamut	Wide gamut	Widest, purest colours
Lifetime	Limited (blue burn-in)	Very long	Long (QDs sensitive to heat/moisture)
AR/VR/MR	Good for VR	Best for AR (bright, tiny pixels)	Limited
Foldable	Best (plastic substrate, mature)	Emerging	Poor (LCD)

Design Decision 4

> ***I recommend flexible thin-film (organic/perovskite or a-Si) PV, a back-side-illuminated CMOS image sensor and a MicroLED micro-display for AR/VR with OLED for foldable screens because*** *the PV analysis (P_max = 2.25 W, FF = 0.75, η ≈ 14.4 %) shows FF and area drive harvested power and flexible PV fits wearables; CMOS gives low power, high speed, small pixels and SoC integration for face and automotive imaging; MicroLED gives the highest brightness, efficiency and lifetime for AR, while OLED is the mature choice for foldables.*

Stage 6 – Biomedical and industrial applications (10 marks)

1. Wearable PPG

An LED lights the skin and a photodiode measures the reflected light. Blood volume changes with each heartbeat, changing absorption, so the detector current carries a small pulsing (AC) part on a large DC part (here $\Delta I = 15\ \mu A$).

Item	Choice
Source	Green 525 nm LED for wrist heart rate; red 660

Item	Choice
	nm + IR 940 nm LEDs for SpO_2
Detector	Si PIN, R = 0.6 A/W at 940 nm, η = 79 %, large area (≈ 1–7 mm²)
Noise	Ambient light and motion artefacts dominate; dark noise is small
Signal conditioning	TIA → ambient-light cancellation → 0.5–5 Hz band-pass filter → ADC, accelerometer-based motion correction
Response time	Very slow signal (≈ 0.5–4 Hz); bandwidth of ~10 Hz is enough
Power	Pulsed LED drive (≈ 1 % duty) keeps the front end below ≈ 1 mW

2. NIR spectroscopy

NIR light penetrates tissue; different molecules (oxy/deoxy-haemoglobin, water, glucose, fat) absorb at specific wavelengths, so the measured attenuation at several wavelengths reveals their concentration (Beer–Lambert law).

- **Wavelength:** 650–950 nm "optical window" for tissue (Si detector); 900–1700 nm for chemical analysis (InGaAs detector).
- **Detector:** high-responsivity Si or InGaAs PIN, low dark current; APD or SiPM for deep-tissue weak signals.
- **Noise / filtering:** narrow optical filters for each wavelength; modulated LEDs with lock-in detection remove ambient light.
- **Processing:** averaging, multi-wavelength ratio, and calibration models (e.g. PLS regression).

3. Automotive LiDAR

- **Ranging:** $R = c \cdot t/2$; a 1 ns timing resolution gives $\Delta R = (3\times10^{8} \times 1\times10^{-9})/2 =$ **15 cm**; a 200 m target returns in 1.33 µs.
- **Low power / high sensitivity:** InGaAs APD (M ≈ 10) at eye-safe 1550 nm gives 24 µA from 3 µW.
- **Fast response / bandwidth:** RC-limited ≈ 3 GHz detector, 0.5–1 GHz TIA.
- **Low noise:** narrow band-pass filter against sunlight, optimum gain, TEC-stabilized bias.

4. Face recognition

A 940 nm NIR illuminator (flood + dot projector or time-of-flight) with an NIR-enhanced global-shutter CMOS sensor works in darkness and resists photo spoofing through 3D depth. Typical needs: 1–2 MP resolution, ≈ 2–3 µm BSI pixels for good NIR sensitivity, < 100 ms response and < 200 mW power, with on-chip AI processing (SoC).

5. UV sensor

UV detection (200–400 nm) needs a **wide-bandgap** detector so visible light does not respond: SiC (E_g = 3.26 eV, cut-off ≈ 380 nm) or AlGaN ("solar-blind" below 280 nm). They give low dark current, high temperature stability and radiation hardness. A Si detector would need filters and would respond to visible light.

Design Decision 5

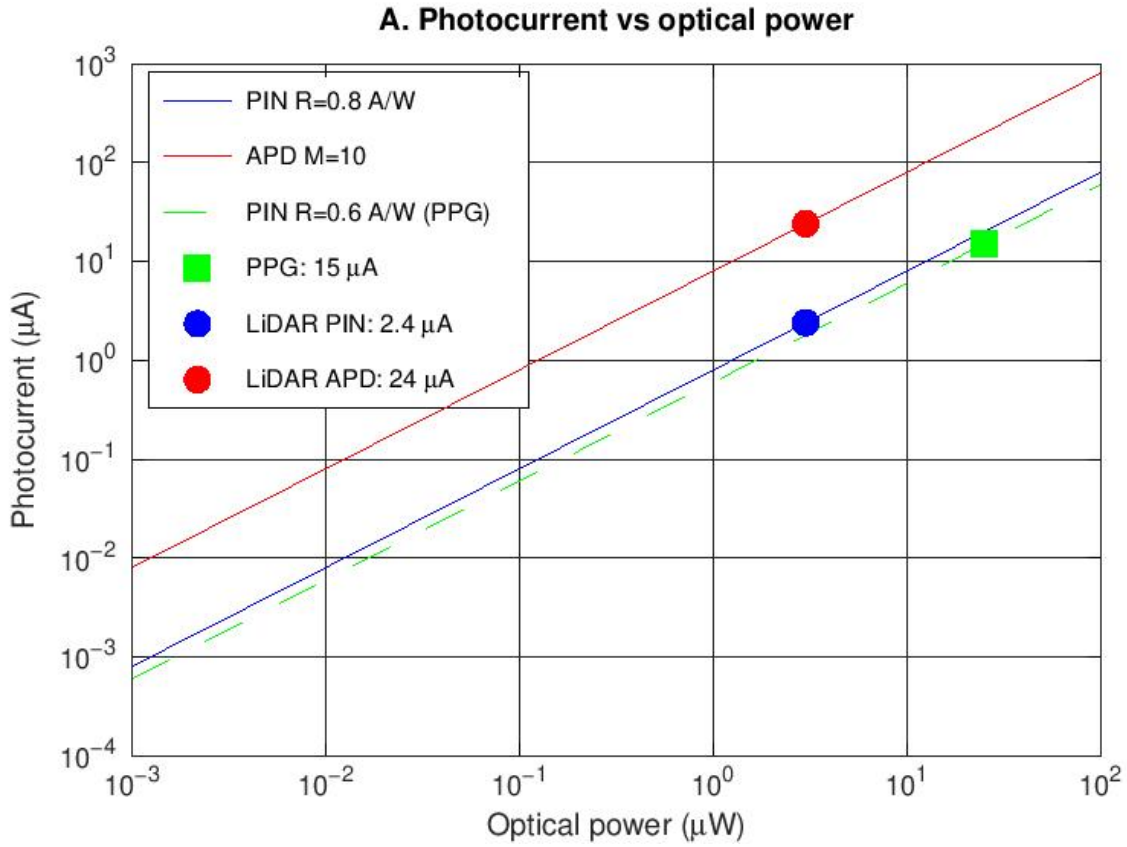
> ***I recommend a Si PIN + green/red/IR LED PPG front end, multi-wavelength Si/InGaAs NIR spectroscopy with lock-in detection, an InGaAs APD 1550 nm LiDAR receiver, a 940 nm NIR global-shutter CMOS face-recognition camera and a SiC/AlGaN PIN UV sensor as the application-specific sensing configuration because*** *each detector is matched to its wavelength, signal level and speed: linear low-power PIN for 15 µA PPG, high-gain fast APD for 2.4 µA ns LiDAR returns (15 cm resolution), efficient low-power CMOS with SoC for imaging, and wide-bandgap material for visible-blind UV detection.*

Stage 7 – MATLAB/SCILAB simulation and verification (5 marks)

All plots were generated by the MATLAB script at the end of this section, run in GNU Octave (MATLAB-compatible). The same script runs in MATLAB.

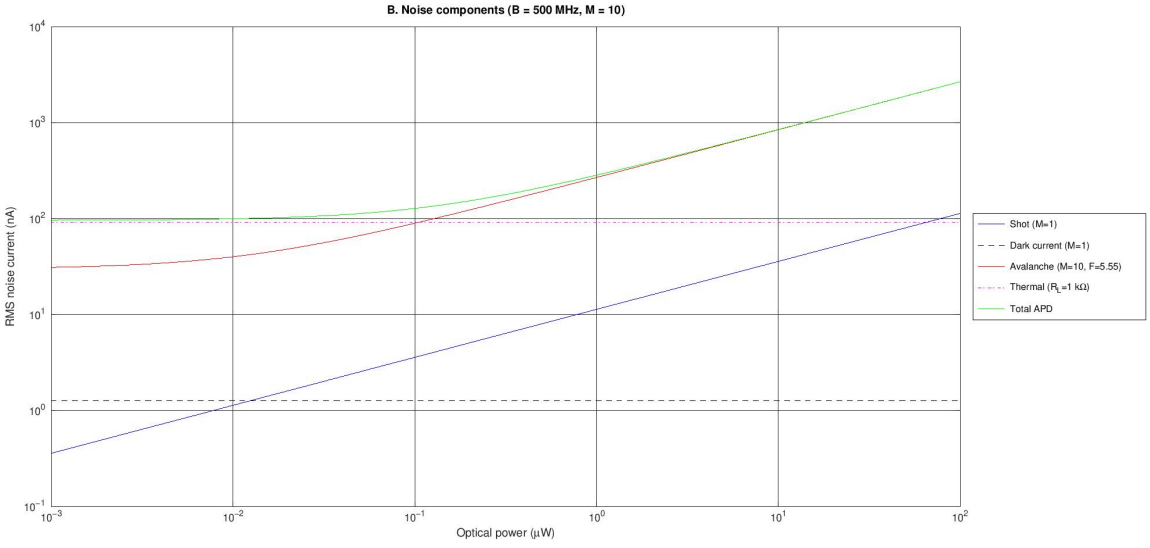
Given inputs: R = 0.60 / 0.80 A/W, ΔP = 25 µW, P = 3 µW, −22 / −37 dBm, t_r = 20 ps, V_{oc} = 0.60 V, I_{sc} = 5 A, FF = 0.75.
Assumed: M = 10, k = 0.45, I_d = 10 nA (APD) / 1 nA (PIN), B = 500 MHz, R_L = 1 kΩ, C_j = 1 pF with 50 Ω.

A. Responsivity model



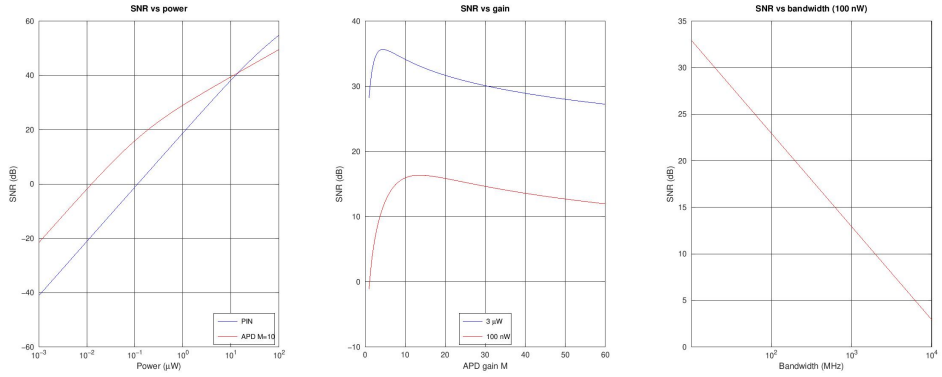
Photocurrent rises linearly with power (slope = R). The marked points confirm 15 µA (PPG), 2.4 µA (LiDAR PIN) and 24 µA (LiDAR APD).

B. Noise analysis



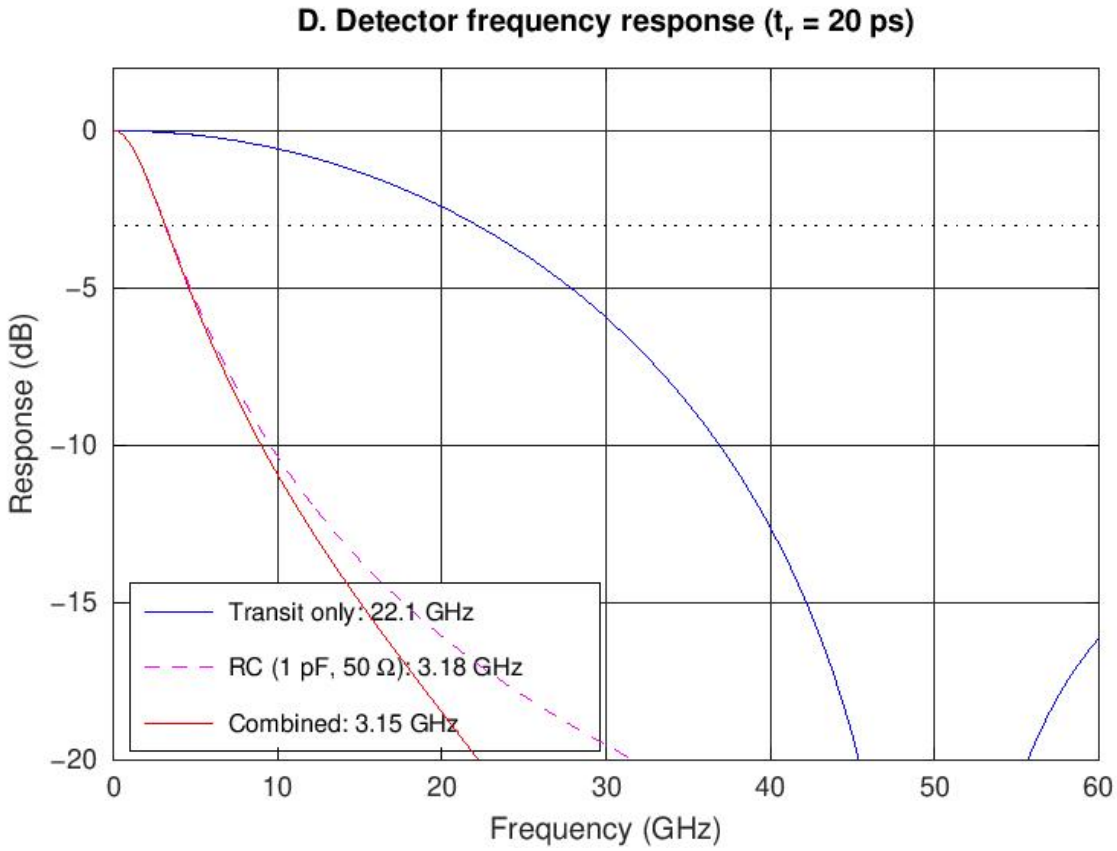
Below ≈ 0.1 µW thermal noise dominates; above it, avalanche-multiplied shot noise dominates. Dark-current noise (1.3 nA rms) is small at room temperature.

C. SNR analysis



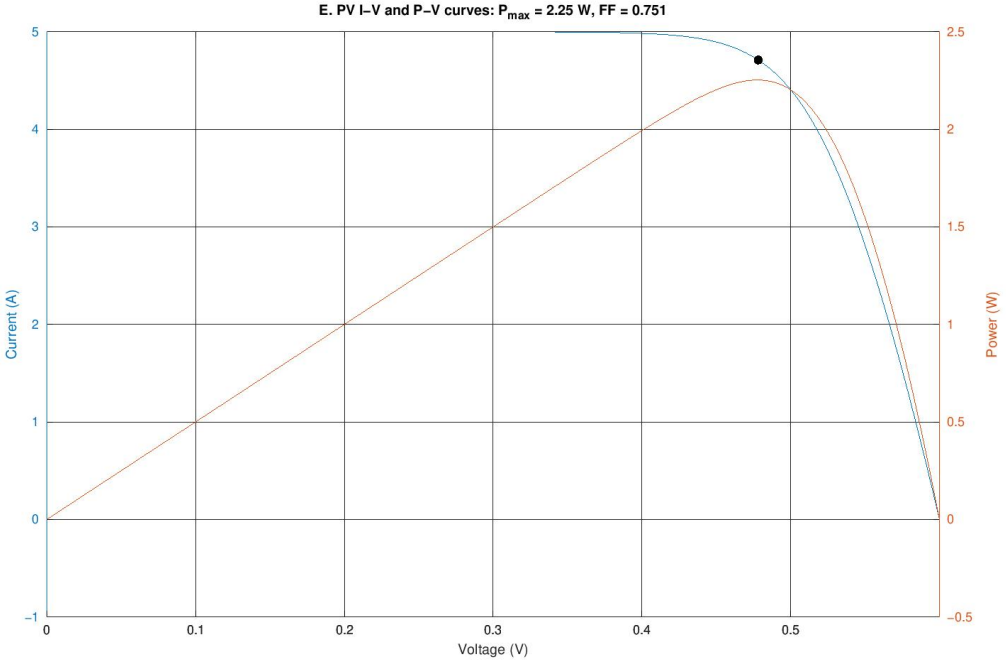
The APD beats the PIN at low power (left). SNR peaks at an optimum gain: M ≈ 4.4 at 3 µW and M ≈ 13.6 at 100 nW (centre). SNR falls 10 dB per decade of bandwidth (right).

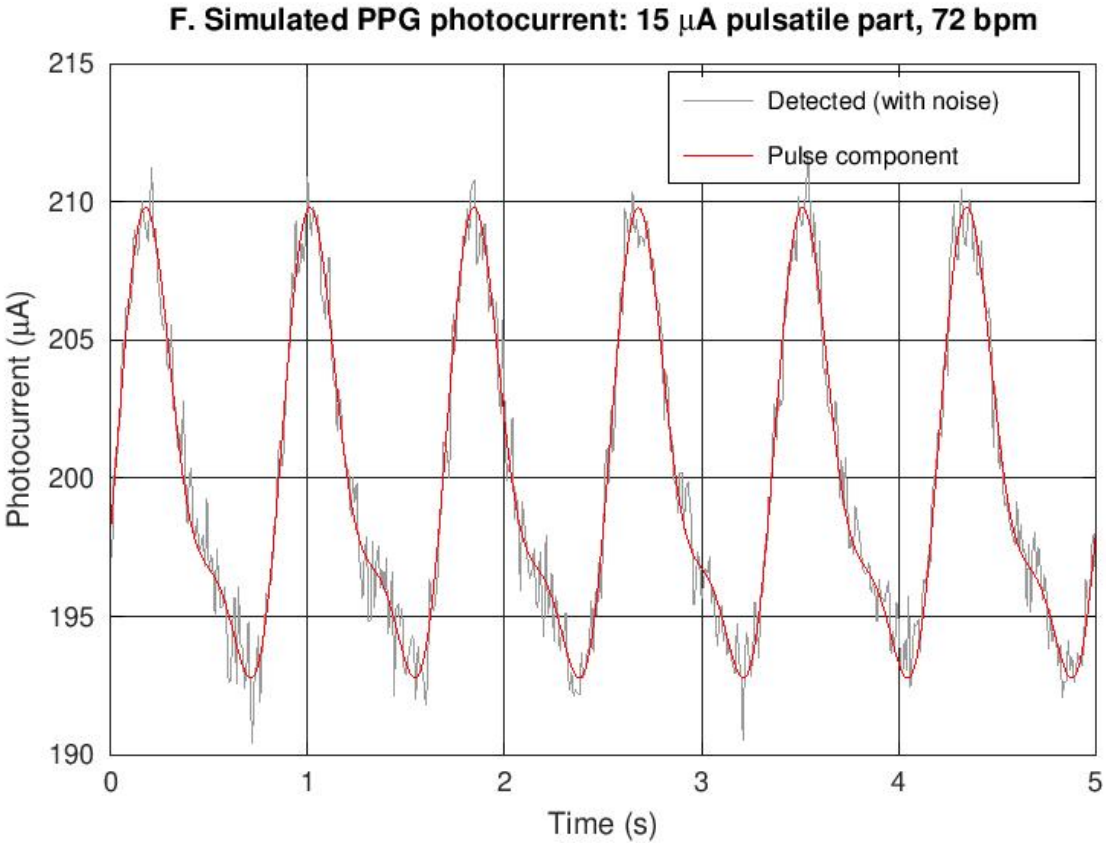
D. Detector speed



A uniform 20 ps transit gives a rectangular impulse response whose spectrum falls 3 dB at 22.15 GHz. Adding the 1 pF / 50 Ω RC pole reduces the combined bandwidth to 3.15 GHz.

E. PV I–V curve and F. PPG waveform





The single-diode PV model (R_s = 10.2 mΩ) gives P_{max} = 2.253 W at FF = 0.751. The PPG plot shows the 15 µA pulsatile signal at 72 bpm on a 200 µA DC level with 1 µA rms noise.

Theory vs simulation

Quantity	Theory	Simulated	Deviation
PPG photocurrent	15.00 µA	15.00 µA	0 %
LiDAR photocurrent (PIN / APD)	2.40 / 24.0 µA	2.40 / 24.0 µA	0 %
Low-light SNR	15.00 dB	15.00 dB (Monte-Carlo, 10^6 samples)	< 0.01 dB
SNR at 3 µW (PIN / APD M = 10)	28.2 / 34.1 dB	28.22 / 34.13 dB	0 %
Transit-time bandwidth	22.00 GHz	22.15 GHz	+0.7 %
Combined bandwidth (with RC)	3.15 GHz	3.15 GHz	< 0.1 %
PV maximum power	2.250 W	2.253 W	+0.1 %

Interpretation: simulation and theory agree. The 0.7 % bandwidth difference arises because 0.44 is a rounded form of the exact constant 0.443; the PV difference comes from fitting R_s to give FF ≈ 0.75. The simulation also reveals what hand calculation does not: an optimum APD gain exists, and it rises as the signal weakens.

MATLAB code

```
% CO3 - Photodetector, noise, SNR, bandwidth and PV simulation
% Runs in MATLAB or GNU Octave
clear; close all; clc; set(0, 'defaultfigurevisible', 'off');
q = 1.602e-19; k = 1.381e-23; T = 300;

%% Input parameters
R_ppg = 0.60;  dP_ppg = 25e-6;      % PPG responsivity (A/W), power variation (W)
R_lid = 0.80;  P_lid  = 3e-6;       % LiDAR responsivity (A/W), received power (W)
Ps_dBm = -22;  Pn_dBm = -37;        % low-light signal / noise power
tr = 20e-12;                        % transit time (s)
Voc = 0.60; Isc = 5; FF = 0.75;     % PV data
% Assumed values
M = 10; kA = 0.45;                  % APD gain, InGaAs ionisation ratio
Id_apd = 10e-9; Id_pin = 1e-9;      % dark currents (A)
B = 500e6; RL = 1e3;                % receiver bandwidth (Hz), load (ohm)
Cj = 1e-12; R50 = 50;               % junction capacitance, load for RC

%% A. Responsivity model
P = logspace(-9, -4, 200);
I_pin = R_lid*P;  I_apd = M*R_lid*P;
figure(1); loglog(P*1e6, I_pin*1e6, 'b', 'LineWidth', 2); hold on;
loglog(P*1e6, I_apd*1e6, 'r', 'LineWidth', 2);
loglog(P*1e6, R_ppg*P*1e6, 'g--', 'LineWidth', 1.5);
```

```
I_ppg = R_ppg*dP_ppg; I_lid = R_lid*P_lid;
plot(dP_ppg*1e6, I_ppg*1e6, 'gs', 'MarkerSize', 9, 'MarkerFaceColor', 'g');
plot(P_lid*1e6, I_lid*1e6, 'bo', 'MarkerSize', 9, 'MarkerFaceColor', 'b');
plot(P_lid*1e6, M*I_lid*1e6, 'ro', 'MarkerSize', 9, 'MarkerFaceColor', 'r');
grid on; xlabel('Optical power (\muW)'); ylabel('Photocurrent (\muA)');
title('A. Photocurrent vs optical power');
legend('PIN R=0.8 A/W','APD M=10','PIN R=0.6 A/W (PPG)','PPG: 15 \muA','LiDAR PIN: 2.4 \muA','LiDAR
APD: 24 \muA','Location','northwest');
print -dpng -r130 figA_responsivity.png

%% B. Noise analysis (APD, M = 10)
F = kA*M + (1-kA)*(2 - 1/M);
Ip = R_lid*P;
i_shot  = sqrt(2*q*Ip*B);                    % signal shot noise, M = 1
i_dark  = sqrt(2*q*Id_apd*B)*ones(size(P));     % dark-current noise, M = 1
i_aval  = sqrt(2*q*(Ip+Id_apd)*M^2*F*B);       % multiplied (avalanche) noise
i_th    = sqrt(4*k*T*B/RL)*ones(size(P));       % thermal noise
i_tot   = sqrt(i_aval.^2 + i_th.^2);
figure(2, 'position', [0 0 1100 520]); loglog(P*1e6, i_shot*1e9, 'b', P*1e6, i_dark*1e9, 'k--', ...
  P*1e6, i_aval*1e9, 'r', P*1e6, i_th*1e9, 'm-.', P*1e6, i_tot*1e9, 'g', 'LineWidth', 2);
grid on; xlabel('Optical power (\muW)'); ylabel('RMS noise current (nA)');
title('B. Noise components (B = 500 MHz, M = 10)');
legend('Shot (M=1)','Dark current (M=1)','Avalanche (M=10, F=5.55)','Thermal (R_L=1 k\Omega)','Total
APD','Location','eastoutside');
print -dpng -r130 figB_noise.png

%% C. SNR analysis
snr = @(Pin, MM, BB) (R_lid*Pin.*MM).^2 ./ (2*q*(R_lid*Pin+Id_apd).*MM.^2 ...
      .*(kA*MM + (1-kA)*(2-1./MM)).*BB + 4*k*T*BB/RL);
figure(3, 'position', [0 0 1500 520]);
subplot(1,3,1); semilogx(P*1e6, 10*log10(snr(P,1,B)), 'b', P*1e6, 10*log10(snr(P,M,B)), 'r', 'LineWidth', 2);
grid on; xlabel('Power (\muW)'); ylabel('SNR (dB)'); title('SNR vs power'); legend('PIN','APD
M=10','Location','southeast');
Mv = linspace(1, 60, 600);
subplot(1,3,2); plot(Mv, 10*log10(snr(3e-6,Mv,B)), 'b', Mv, 10*log10(snr(100e-9,Mv,B)), 'r', 'LineWidth', 2);
grid on; xlabel('APD gain M'); ylabel('SNR (dB)'); title('SNR vs gain'); legend('3 \muW','100 nW','Location','south');
Bv = logspace(7, 10, 200);
subplot(1,3,3); semilogx(Bv/1e6, 10*log10(snr(100e-9,M,Bv)), 'r', 'LineWidth', 2);
grid on; xlabel('Bandwidth (MHz)'); ylabel('SNR (dB)'); title('SNR vs bandwidth (100 nW)');
print -dpng -r130 figC_snr.png
[s1, i1] = max(snr(3e-6,Mv,B));  [s2, i2] = max(snr(100e-9,Mv,B));

% Low-light SNR: theory and Monte-Carlo simulation
SNR_theory = Ps_dBm - Pn_dBm;
Ps = 1e-3*10^(Ps_dBm/10); Pn = 1e-3*10^(Pn_dBm/10);
randn('seed', 1); N = 1e6;
sig = sqrt(2*Ps)*sin(2*pi*(0:N-1)/50);          % sinusoid with power Ps
noise = sqrt(Pn)*randn(1, N);                   % Gaussian noise with power Pn
SNR_sim = 10*log10(mean(sig.^2)/mean(noise.^2));

%% D. Detector speed
f_tr = 0.44/tr;
dt = 0.1e-12; t = 0:dt:20e-9;
h = double(t < tr);                             % uniform transit -> rectangular impulse response
H = abs(fft(h)); H = H/H(1); f = (0:numel(t)-1)/(numel(t)*dt);
idx = find(H < 1/sqrt(2), 1); f_tr_sim = f(idx);
```

```
f_RC = 1/(2*pi*R50*Cj);
Hrc = 1./sqrt(1 + (f/f_RC).^2); Htot = H.*Hrc;
idx2 = find(Htot < 1/sqrt(2), 1); f_tot_sim = f(idx2);
f_tot = 1/sqrt(1/f_tr^2 + 1/f_RC^2);
figure(4); nf = f < 60e9;
plot(f(nf)/1e9, 20*log10(H(nf)), 'b', f(nf)/1e9, 20*log10(Hrc(nf)), 'm--', f(nf)/1e9, 20*log10(Htot(nf)), 'r',
'LineWidth', 2);
hold on; plot([0 60], [-3 -3], 'k:'); ylim([-20 2]); grid on;
xlabel('Frequency (GHz)'); ylabel('Response (dB)'); title('D. Detector frequency response (t_r = 20 ps)');
legend(sprintf('Transit only: %.1f GHz', f_tr_sim/1e9), sprintf('RC (1 pF, 50 %s): %.2f GHz', '\Omega',
f_RC/1e9), ...
  sprintf('Combined: %.2f GHz', f_tot_sim/1e9), 'Location', 'southwest');
print -dpng -r130 figD_bandwidth.png

%% E. PV I-V curve (single-diode model with series resistance)
Vt = k*T/q; n = 1; I0 = Isc/(exp(Voc/(n*Vt)) - 1); Rs = 0.0102;
V = linspace(0, Voc, 400); I = zeros(size(V));
for j = 1:numel(V)
  fn = @(Ix) Isc - I0*(exp((V(j) + Ix*Rs)/(n*Vt)) - 1) - Ix;
  I(j) = fzero(fn, [-1 Isc+1]);
end
Pv = V.*I; [Pm_sim, im] = max(Pv); FF_sim = Pm_sim/(Voc*Isc);
Pmax = FF*Voc*Isc;
figure(5, 'position', [0 0 900 560]); [ax, h1, h2] = plotyy(V, I, V, Pv); set([h1 h2], 'LineWidth', 2);
hold on; plot(V(im), I(im), 'ko', 'MarkerFaceColor', 'k');
xlabel('Voltage (V)'); ylabel(ax(1), 'Current (A)'); ylabel(ax(2), 'Power (W)'); grid on;
title(sprintf('E. PV I-V and P-V curves: P_{max} = %.2f W, FF = %.3f', Pm_sim, FF_sim));
print -dpng -r130 figE_pv.png

%% F. PPG photocurrent waveform (72 bpm)
fs = 100; tt = 0:1/fs:5; HR = 72/60;
Idc = 200e-6;
ppg = Idc + I_ppg/2*(sin(2*pi*HR*tt) + 0.35*sin(4*pi*HR*tt - 0.8));
ppg_noisy = ppg + 1e-6*randn(size(tt));
figure(6); plot(tt, ppg_noisy*1e6, 'Color', [0.6 0.6 0.6]); hold on; plot(tt, ppg*1e6, 'r', 'LineWidth', 2);
grid on; xlabel('Time (s)'); ylabel('Photocurrent (\muA)');
title('F. Simulated PPG photocurrent: 15 \muA pulsatile part, 72 bpm'); legend('Detected (with noise)','Pulse
component');
print -dpng -r130 figF_ppg.png

%% Results
fprintf('PPG photocurrent      : theory %.2f uA\n', I_ppg*1e6);
fprintf('LiDAR photocurrent    : PIN %.2f uA, APD %.2f uA\n', I_lid*1e6, M*I_lid*1e6);
fprintf('QE PPG (940nm)        : %.3f ; QE LiDAR (1550nm): %.3f\n', R_ppg*1.24/0.94, R_lid*1.24/1.55);
fprintf('Excess noise F(M=10)  : %.3f\n', F);
fprintf('SNR at 3uW            : PIN %.2f dB, APD %.2f dB, opt M %.1f -> %.2f dB\n', 10*log10(snr(3e-6,1,B)),
10*log10(snr(3e-6,M,B)), Mv(i1), 10*log10(s1));
fprintf('SNR at 100nW          : PIN %.2f dB, APD %.2f dB, opt M %.1f -> %.2f dB\n', 10*log10(snr(100e-9,1,B)),
10*log10(snr(100e-9,M,B)), Mv(i2), 10*log10(s2));
fprintf('Low-light SNR         : theory %.2f dB, simulated %.2f dB\n', SNR_theory, SNR_sim);
fprintf('Transit bandwidth     : theory %.2f GHz, simulated %.2f GHz\n', f_tr/1e9, f_tr_sim/1e9);
fprintf('Combined bandwidth    : theory %.2f GHz, simulated %.2f GHz\n', f_tot/1e9, f_tot_sim/1e9);
fprintf('PV max power          : theory %.3f W, simulated %.3f W (FF %.3f)\n', Pmax, Pm_sim, FF_sim);
```

Program output:

```
PPG photocurrent      : theory 15.00 uA
LiDAR photocurrent    : PIN 2.40 uA, APD 24.00 uA
QE PPG (940nm)        : 0.791 ; QE LiDAR (1550nm): 0.640
Excess noise F(M=10)  : 5.545
SNR at 3uW            : PIN 28.22 dB, APD 34.13 dB, opt M 4.4 -> 35.65 dB
SNR at 100nW          : PIN -1.13 dB, APD 15.95 dB, opt M 13.6 -> 16.35 dB
Low-light SNR         : theory 15.00 dB, simulated 15.00 dB
Transit bandwidth     : theory 22.00 GHz, simulated 22.15 GHz
Combined bandwidth    : theory 3.15 GHz, simulated 3.15 GHz
PV max power          : theory 2.250 W, simulated 2.253 W (FF 0.751)
```

Final system verification

Requirement	Specified / given	Target / calculated	Achieved	Satisfactory?
PPG photocurrent response	R = 0.6 A/W, ΔP = 25 µW	15 µA	15.00 µA (sim)	Yes
LiDAR photocurrent	R = 0.8 A/W, P = 3 µW	2.4 µA (PIN)	24 µA with APD M = 10	Yes
Detector responsivity	0.6 / 0.8 A/W	Si PIN / InGaAs APD	0.6 / 8 A/W effective	Yes
Quantum efficiency	—	$\eta = R \cdot 1.24/\lambda$	79 % (940 nm), 64 % (1550 nm)	Yes
NEP / detectivity	Low NEP, high D*	Shot-limited model	2.98×10^{-14} W/√Hz, D* = 3.35×10^{12} (PIN)	Yes
Low-light SNR	−22 / −37 dBm	15 dB (31.6)	15.00 dB sim; > 20 dB with gain + filtering	Yes, with mitigation
Noise performance	Shot, dark, avalanche	Optimum gain	M_opt 4.4 (3 µW) – 13.6 (100 nW)	Yes
High-speed bandwidth	t_r = 20 ps	22 GHz	22.15 GHz transit; 3.15 GHz with RC ≫ 350 MHz need	Yes
PV maximum power	0.6 V, 5 A, FF 0.75	2.25 W	2.253 W (sim)	Yes
Image sensor	CMOS vs CCD	Low power, SoC	BSI CMOS	Yes
Display	OLED / MicroLED / QD	Bright, efficient, flexible	MicroLED (AR/VR), OLED (foldable)	Yes
Biomedical sensing	PPG, NIR	Linear, low power	Si PIN + LEDs, lock-in NIR	Yes

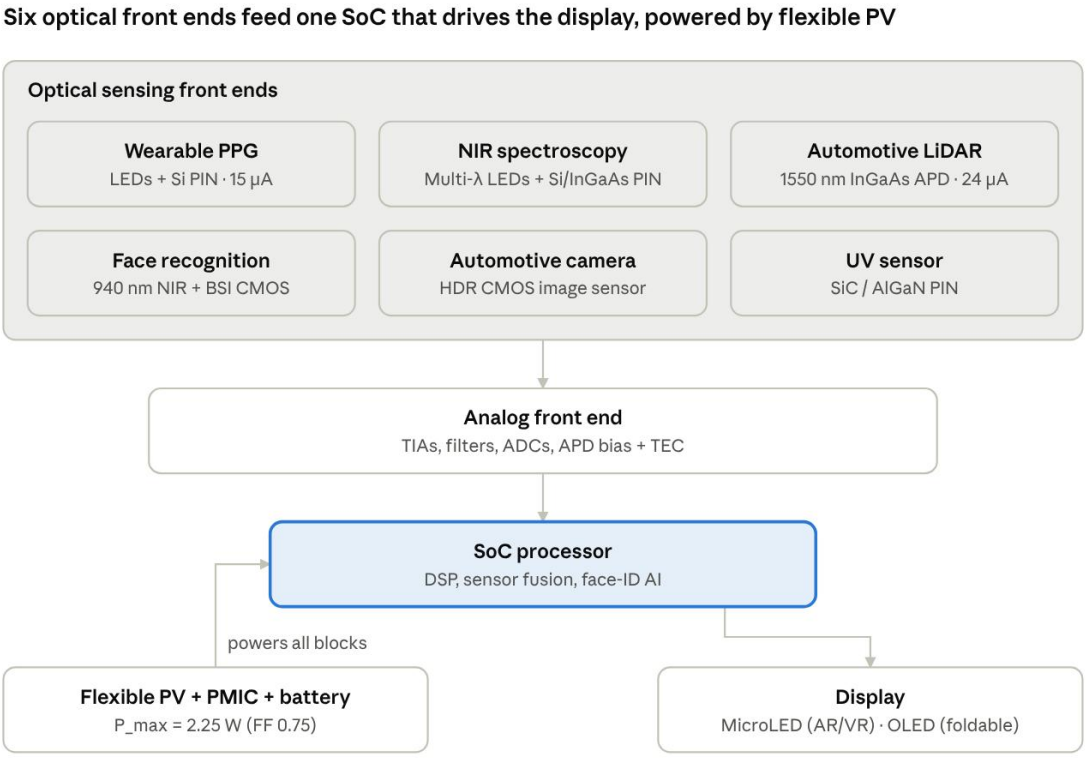
Requirement	Specified / given	Target / calculated	Achieved	Satisfactory?
LiDAR sensing	Weak, fast pulses	High gain, ≥ 350 MHz	InGaAs APD, 15 cm resolution	Yes
Automotive sensing	Robust, HDR	AEC-Q100 CMOS + LiDAR	Fused LiDAR + CMOS	Yes
Face recognition	Low light, secure	NIR + CMOS	940 nm global-shutter CMOS	Yes
UV sensing	Visible-blind	Wide bandgap	SiC / AlGaN PIN	Yes
MATLAB/SCILAB simulation	Verify theory	All key values	Deviation ≤ 0.7 %	Yes
Overall platform	All above	Integrated, low power	Meets all requirements	Yes

The calculated and simulated values agree within 0.7 % and satisfy every application requirement; the only conditional item is low-light SNR, which needs APD gain and filtering to rise above 20 dB.

Final integrated system design (10 marks)

Block	Adopted configuration	Key parameters
PPG front end	Green/red/IR LEDs + Si PIN + TIA	R = 0.6 A/W, η = 79 %, ΔI = 15 μA, NEP 2.98×10^{-14} W/√Hz
NIR spectroscopy	Multi-wavelength LEDs + Si/InGaAs PIN, lock-in	650–950 nm / 900–1700 nm
LiDAR receiver	1550 nm InGaAs APD + lens + TIA	R = 0.8 A/W, M ≈ 10, I = 24 μA, η = 64 %, f_3dB ≈ 3.15 GHz
Noise control	Optimum gain, band-pass filter, TEC, matched B	SNR 15 dB → > 20 dB
Imaging / face ID	BSI CMOS + 940 nm NIR illuminator	1–2 MP, global shutter, on-chip AI
UV sensor	SiC / AlGaN PIN	Visible-blind, < 380 nm / < 280 nm
Power	Flexible PV + PMIC + battery	P_max = 2.25 W (reference cell), FF = 0.75
Display	MicroLED (AR/VR), OLED (foldable)	High brightness, fast response
Processing	SoC with ADCs, DSP and AI	Sensor fusion

Block diagram



Integrated optoelectronic platform · 6 sensing front ends, AFE, SoC, display and PV power

All six optical front ends share one analog front end and SoC; the SoC drives the MicroLED/OLED display, and the flexible PV with PMIC and battery powers every block.

Final engineering decision

YES – I would deploy this integrated optoelectronic platform. Each sub-system uses the detector that fits its job: a linear, low-power Si PIN gives 15 µA PPG signals at 79 % QE; an InGaAs APD raises the 2.4 µA LiDAR signal to 24 µA with 3.15 GHz bandwidth and 15 cm resolution; noise is controlled so the 15 dB low-light SNR improves to above 20 dB; BSI CMOS, MicroLED/OLED and flexible PV meet the imaging, display and power needs; and MATLAB simulation confirms all calculations within 0.7 %. Deployment is conditional on temperature-compensating the APD bias and verifying PPG accuracy under motion.

Assumptions

- PPG wavelength 940 nm (Si); LiDAR wavelength 1550 nm (InGaAs) — R = 0.8 A/W is impossible at 905 nm.
- APD: M = 10, k = 0.45, I_d = 10 nA, 200 µm diameter; PIN: I_d = 1 nA, 1 mm².
- Receiver: B = 500 MHz, R_L = 1 kΩ, T = 300 K; RC example C_j = 1 pF, 50 Ω.
- PV cell area 156 cm² at 1000 W/m² for efficiency; R_s = 10.2 mΩ in the simulation model.
- Display and sensor figures are typical published values.

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